INSTITUT PASTEUR

Microbes and Infection 13 (2011) 134–142

www.elsevier.com/locate/micinf

Review

Cell-mediated immunity in human brucellosis

Panagiotis Skendros [a,*], Georgios Pappas [b], Panagiota Boura [c]

[a] First Department of Internal Medicine, Democritus University of Thrace, Alexandroupolis, Greece
[b] Institute of Continuing Medical Education of Ioannina, Greece
[c] Clinical Immunology Unit, 2nd Department of Internal Medicine, Aristotle University of Thessaloniki, Thessaloniki, Greece

Received 2 September 2010; accepted 15 October 2010
Available online 27 October 2010

Abstract

Brucella can parasitize within human antigen-presenting cells modifying phagocytosis, phagolysosome fusion, antigen presentation, cytokine secretion, and apoptosis. Subversion of innate immune mechanisms by *Brucella* leads to defective Th1 immune responses and T-cell anergy in chronic brucellosis patients. This review summarizes the cellular immune responses in brucellosis, based on data derived exclusively from human cells or cell lines.
© 2010 Institut Pasteur. Published by Elsevier Masson SAS. All rights reserved.

Keywords: Immunology; Cellular immunity; *Brucella*; Human brucellosis

1. Introduction

Brucellosis is the commonest bacterial zoonotic infection, annually affecting in a worldwide distribution more than half a million individuals, an incidence that is definitely an underestimate due to extended under-reporting in several endemic countries. In recent years, new disease foci emerge or re-emerge globally, altering the epidemiology of the disease and enhancing the role of *Brucella* in travel and immigration-related infections [1]. The burden of the disease is not likely to minimize in the near future, since there exists no efficient vaccine for humans, and disease prevention depends on complex and costly animal control and eradication projects [1].

Brucellosis is caused by intracellular pathogens of the genus *Brucella*, the natural reservoir of which is domestic and wild animals. *Brucella melitensis*, *Brucella abortus* and *Brucella suis* induce human disease; rare but persisting cases of *Brucella canis* human infection are also recognized, as well as human disease by novel *Brucella* species of marine mammals [1]. *Brucella* species infect humans by consumption of contaminated dairy products or by occupational contact with infected animals, but also through aerosol transmission due to its volatile properties and low bacterial inoculum needed to induce infection [2]: thus *Brucella* is considered as one of the most common laboratory-acquired pathogens [3] and is furthermore classified as category B agent on the Center for Disease Control biodefense list.

In humans the disease causes high clinical morbidity and protean clinical manifestations, as any organ can be affected [1]. Despite early diagnosis and treatment, approximately 10–30% of patients develop chronic disease, characterized by atypical clinical presentations, chronic fatigue syndrome and relapses [4]. It is estimated that about 10% of patients treated with the varying existing therapeutic regimens proceed to a chronic form of brucellosis [1].

The clarification and better understanding of the immunopathogenic mechanisms evolving in human brucellosis is important for the identification of markers for disease severity, chronicity, and treatment response as well as the development of new therapeutic/preventive strategies.

Brucella interplay with the human immune system is critical for the development of chronic parasitism or infection clearance [5,6]. Host protection against *Brucella* depends on cell-mediated immunity, involving mainly activated antigen-presenting cells (macrophages, dendritic cells) and CD4+, CD8+

* Corresponding author.
E-mail address: pskendro@med.duth.gr (P. Skendros).

1286-4579/$ - see front matter © 2010 Institut Pasteur. Published by Elsevier Masson SAS. All rights reserved.
doi:10.1016/j.micinf.2010.10.015

T-lymphocytes [5,7]. *Brucella* antigens induce the production of T-helper 1 (Th1) cytokines in humans; thus Th1 immune response is essential for the clearance of *Brucella* infection [8]. *Brucella* on the other end can survive and replicate in host macrophages and dendritic cells subverting innate immunity and evading adaptive immune mechanisms [7,9].

In the majority of the studies concerning immunity against *Brucella*, cells or cells lines derived from domestic ruminants or mice were used. However, brucellosis pathophysiology in ruminants differs from that of humans. Moreover, mice are characterized by natural resistance to *Brucella* infection and do not develop disease comparable to human brucellosis. In accordance to this, it is important to note that immune mechanisms against *Brucella* and their significance may diverge among different hosts [6,9]. For example, human and mice antigen-presenting cells, such as dendritic cells, display several differences leading to variations in immune response. From this point of view, this review summarizes the cellular immune responses in human brucellosis, emphasizing on the chronic form of the disease, and based on data derived exclusively from human cell lines/cells.

2. Brucella and human professional antigen-presenting cells cross talk

2.1. Macrophages

Macrophages are thought to be key elements in the cellular immune response against intracellular bacteria like *Brucella*. Activated macrophages juxtapose a wide array of phagocytic and inducible bactericidal functions to kill *Brucella*, like degradative hydrolytic enzymes concentrated within lysosomal compartment, acidification of phagolysosomes, antimicrobial cationic peptides (defensins), nutritional deprivation and oxidative burst (reactive oxygen and nitrogen species) [6,10].

Moreover, infected macrophages produce pro-inflammatory cytokines (TNFα, IL-6, IL-12) and chemokines (GRO-α, IL-8, MCP-1, RANTES, MIP1α/MIP1β) [11,12]. Among these biological mediators the role of TNFα and IL-12 is critical. TNFα strongly enhances the bactericidal activity of phagocytes and IL-12 drives the Th1 immune response, leading to the production of IFNγ, the principal cytokine fighting *Brucella* infection [5,6,8]. As mentioned below, the inhibition of TNFα production by live *Brucella* is used as a strategy for intracellular survival and establishment of chronic parasitism [13]. In clinical practice, brucellosis reactivation and complications have been reported in rheumatology patients under anti-TNFα biological treatment [14].

Despite the above effector mechanisms, *Brucella* can survive and replicate for prolonged periods within host macrophages of the reticuloendothelial system producing chronic, and even lifelong, infections [15]. To achieve this, *Brucella* produces various virulence factors that modify phagocytosis, phagolysosome fusion, antigen presentation, cytokine secretion, and apoptosis.

Studies on human monocytes, monocyte-derived macrophages and human cell lines indicate that the microbe is uptaken via conventional phagocytosis using various phagocytosis-promoting receptors. Following entry, 80–90% of bacteria are killed within the first few hours, however some bacteria resist macrophage killing and are able to survive and replicate intracellularly by remodeling their phagosomes and preventing phagolysosomal fusion [16,17].

More specifically, studies on cultured human peripheral blood monocytes, demonstrated that regular spacious phagosomes represent the killing compartment, whereas special tight phagosomes (named as *Brucella* containing vacuoles or BCV) the survival-permitting compartment. Although BCV avoid fusion with lysosomes in these host cells, no association was detected between BCV and the rough endoplasmic reticulum (ER) or autophagosomes [16]. Both the engulfment and intracellular localization are independent of phagocytosis-promoting receptors (opsonic/nonopsonic) or species-specific differences [16]. In contrast, another study examining the model of *B. abortus* chronic intracellular infection in human monocytic cell line THP-1 showed that nonopsonized bacteria promote ER positive BCV. Nevertheless both opsonized and nonopsonized bacteria are equally capable of replicating in the above model [17]. Contradictory results are also reported concerning the pH of BCV. In cultured primary human monocytes survival of internalized *Brucella* depends on an acidic intraphagosomal pH, whereas in THP-1 cell line opsonized *Brucella* replicates within nonacid phagosomes [16,17]. Possibly *Brucella* is acting in a dual manner in order to survive and replicate: resists intraphagosomal acidification or/and induces the deacidification of BCV.

The virB operon of *Brucella* encoding the type IV secretion system (T4SS) is a membrane-associated transporter used to deliver substrate molecules to target cells [18]. T4SS is crucial for the development of the BCV in the host cells as it has been described to modify the intracellular trafficking of the bacteria. Interestingly, *Brucella* induces the expression of the virB operon when exposed to environmental stresses, such as acidic pH and nutritional deprivation, during its intracellular parasitism in host macrophages [18]. The effector proteins transported by the T4SS and the way they influence trafficking patterns of BCV in human macrophages are still unknown, but the importance of the *Brucella* type IV secretion machinery has been established since virB mutant strains fail to replicate in cultured primary and THP-1 human macrophages in comparison to parental strains [18].

Brucella virulence factor A (BvfA) is a small protein induced within human THP-1-derived macrophages by phagosome acidification and is co-regulated with the *Brucella* virB operon [19]. According to this, a role in the establishment of the intracellular replication niche, perhaps as a part of a complex secreted from the periplasm by the *Brucella* T4SS, has been speculated [19]. However, whether or not BvfA is a substrate for the T4SS has not been elucidated, and the function of this protein is unknown yet.

Brucella external membrane components have also been described as virulence factors. Among them *Brucella* lipopolysaccharide (LPS) is considered to be critical for survival and replication in the host [20]. It is an unconventional, non-endotoxic LPS, as compared to classical LPS from

Table 1
Brucella modulation and evasion mechanisms in human professional antigen-presenting cells.

- Prevention of phagocytosis (MMs) [12,21]
- Inhibition of phagolysosome fusion (MMs) [12,16,17]
- Inhibition of cellular maturation (DCs) [31,32]
- Down regulation of antigen presentation (MMs, DCs) [26,32]
- Prevention of apoptosis (MMs) [21,44]
- Blockade of TNFα (MMs, DCs) and IL-12 production (DCs) [13,25,32]
- Interference with IFNγ signal transduction (MMs) [29]

MMs = macrophages, DCs = dendritic cells.

enterobacteria such as *Escherichia coli*. Human monocytes challenged by *Brucella* LPS release lower amounts of IL-1β and TNFα compared to *E. coli* LPS. This effect may be related to the expression of O-polysaccharides (OPS) of the *Brucella* LPS, given that smooth LPS (S-LPS) strains (wild type) elicit reduced levels of IL-6 and TNFα by human monocytes, in comparison to rough mutant strains [12] (On the other hand, one may presume that the non-expression of OPS should allow the bacteria to expose more endotoxic molecules such pro-inflammatory outer membrane proteins). From the clinical point of view, human brucellosis patients do not develop severe sepsis or septic shock such as patients with systemic infections caused by gram-negative bacteria possessing a highly endotoxic LPS. Additionally to the attenuated endotoxic response, *Brucella* S-LPS acts as a potent immunomodulating agent. Interestingly, in human monocytes/macrophages, only S-LPS *Brucella* strains inhibit phagocytosis, bactericidal function and prevent host cells apoptosis forming a proper viable environment to replicate and parasitize [12,21]. In addition, S-LPS protects *Brucella* from complement attack [22] and is able to induce IL-10, a major anti-Th1 cytokine, by human peripheral blood mononuclear cells (PBMCs) [23]. These observations collectively demonstrate that *Brucella* S-LPS is a pathogen-associated molecular pattern (PAMP) that acts in a dual strategy: at early stages of infection elicits a low immunostimulatory activity "masking" the microbe from innate immune response and subsequently modulates the innate immune mechanisms to favor pathogen intracellular localization and replication.

Beyond S-LPS, other PAMPs, such as outer membrane proteins/lipoproteins (Omps) of *Brucella*, are also considered important determinants of human anti-brucellar immunity [24]. In that respect, *Brucella* lipoproteins, and not its LPS, have been suggested as responsible molecules for the activation of monocytes/macrophages and the induction of the pro-inflammatory response elicited by heat-killed *B. abortus* (HKBA) [24]. Previous studies have shown that, in contrast to mice, live *Brucella* spp failed to induce TNFα release upon infection of human macrophages [13], and *Brucella* Omp25 porin is a negative regulator of TNFα production [25]. Recently researchers have demonstrated that *Brucella* lipoprotein Omp19 down-modulates the IFNγ induced MHC-II expression and antigen presentation of human monocytes [26].

Table 2
Disturbances of T-cell immunity in chronic persistent/relapsing brucellosis patients.

- Decline in skin reactivity to different bacterial antigens [42,43]
- Low lymphoproliferation response to mitogens or specific antigens [49–52]
- Diminished production of Th1 cytokines (IFNγ, IL-2) [49,50]
- Increase of CD3+/CD8+ T-lymphocytes [47,48]
- Reduction of CD69+ T-lymphocytes [47]
- Reduction of CD4+/CD25+ T-lymphocytes [48,53]
- Reduction of CD3+/IFNγ+ T-cells [50]
- Increase of CD3+/IL-13+ T-cells (Th2 switch) [50]
- Increased levels of TGFβ1 [51]

Toll-like receptors (TLRs) expressed on antigen-presenting cells serve as essential pathogen recognition receptors (PRRs) with central roles in induction of innate immune responses as well as the subsequent development of adaptive immune responses. Remarkably, *Brucella* spp genomes encode a protein, named TcpB that bears significant homology with mammalian Toll/IL-1 receptor (TIR) domains. TcpB was shown to impede TLR2 and TLR4-mediated signaling in human embryonic kidney (HEK) cells blocking MyD88-induced activation of NF-kB. More precisely, TcpB protein expression interferes with the TLR2/TLR4 adapter MyD88-adapter-like (MAL)/TIRAP representing a pathogen-derived molecule that suppresses host innate-immune responses by specifically targeting the TLR signaling pathway [27,28]. It seems that, in addition to S-LPS and Omps, TcpB may help *Brucella* avoid recognition by innate immune cells by decreasing the TLR-dependent response to infection.

Brucella infected macrophages are resistant to IFNγ activation but the responsible PAMPs and the precise pathogenetic mechanism have not been yet elucidated. Recently, Bouhet et al. reported that *Brucella* escapes the IFNγ-initiated microbicidal activity of human macrophages in a VD3-differentiated THP-1 cell model [29]: The crucial step is likely an interference with IFNγ signal transduction in infected macrophages through marked reduction of IFNγ-mediated association of STAT1 protein with CBP/P300 transactivators. It is suggested that *Brucella* probably triggers the host cell cyclic adenosine monophosphate/protein kinase A (cAMP/PKA) pathway during macrophage infection [29]. Cyclic AMP is a key intracellular second messenger, which at increased levels, has been shown to have anti-inflammatory and anti-apoptotic effects. Entry of *Brucella* in THP-1-derived macrophages was associated with activation of cAMP/PKA pathway required for the bacteria development [30]. In this regard, activation of cAMP/PKA pathway could be responsible for the regulation of various mechanisms that favor *Brucella* infection in human macrophages, such as TNFα/IFNγ production and apoptosis [9].

2.2. Dendritic cells

Myeloid dendritic cells (DCs) display common ontogeny with macrophages and serve as sentinels for the immune system. They play a cardinal role in host defense by ingesting pathogens at the site of infection and migrating to secondary lymphoid organs, where they present pathogen-derived peptides to naive T-lymphocytes, thus priming adaptive immunity.

The immune role of DCs against *Brucella* has been subject of enhanced investigation in recent years. It has been established that *Brucella* spp efficiently invade human monocyte-derived

Table 3
Immunogenetic studies concerning human brucellosis.

Target gene	Pts/controls	*p*	Origin	Reference
TNFα (-308) AG, (-238) **HLA** DRB1, DQB1	59/160	s, ns ns, ns	Spain	[61]
CCR5 Δ32 mutation	185/141	ns	Greece	[62]
NOS2 TAAAn and CCTTTn microsatellites	85/100	ns	Spain	[63]
IFNγ (+874) AA **IL-10** (-1082, -819, -592)	83/101	s ns	Spain	[64]
HLA B39	57/73	s*	Spain	[65]
NRAMP1 [5′(GT)n, −236 CT, D543N, 32UTR]	65/89	ns	Spain	[66]
TLR4 (Asp299Gly) G, AG	198/111	s (s^), s (s^)	Iran	[67]
IL-4 (-590) T, CT, CC	282/163	s, s, r	Iran	[68]
CD14 (-159) TT, CC	228/129	s (s*), r	Iran	[69]
L-selectin (Phe206Leu), allele 206Leu	374/245	s (s$)	Iran	[70]
IL-1 receptor antagonist (intron 2), allele 2	255/162	s	Iran	[71]
TNFα (-308) GG, (-238) GG **IFNγ** (+874) **IL-2** (-330), (+166)	47/166	s, s ns ns, ns	Iran	[72]
TGFβ1 (codons 10/25) 10 CC, 25 GG TC/GG high producers 10 TT	425/213	 s s r	Iran	[73]
IL-10 (-1082, -819, -592) GCC/ATA, high/intermediate producers **IL-6** (-174) GC, high producers **TGFβ1** (codons 10/25) TC/GC **IFNγ** (+874) **TNFα** (-308)	40/50	s s s ns ns	Turkey	[74]
IFNγ (+874) AA **IL-4** (-590) CC IFNγT/IL-4C IFNγTT/IL-4CC	195/91	s r r r	Iran	[75]
MICA A4, A5	114/121	r, s[f]	Spain	[76]
E-selectin (Ser128Arg), ArgArg	258/193	s (s[#])	Iran	[77]
FcγRIIα H131, R131 alleles	150/125	ns	Iran	[78]
IL-10 (-1082) (-819) C, (-592) C (-819) TT, (-592) AA (-1082, -819, -592) ATA/ATA	190/81	ns s, s r, r r	Iran	[79]
TGFβ1 (codons 10/25) TT/GG, TC/GG, 10 CC **IL-6** (-174)	82/102	s, r, s[f] ns	Spain	[80]
IL-6 (-174) GC, GG **IL-10** (-1082), (-819) CT, (-819) CC **IFNγ** (+874) TT, AT, AA **TGFβ1** (codons 10/25) 10 CC, 25 GG **TNFα** (-308)	85/85	s[f], s[nf] ns, s, r s, r, r s, s ns	Turkey	[81]
PTPN22 1858CT	111/150	ns	Spain	[82]
LMP2, LMP7, TAP1, TAP2	61/102	ns	Spain	[83]

pts = patients, s = significant susceptibility, r = significant resistance, ns = nonsignificant, * = osteoarticular complications, ^ = males, $ = relapse, [f] = focal forms, [nf] = non focal forms, # = ≤38 weeks disease duration.

DCs, grow extensively within them [31], and furthermore prevent human DCs maturation and antigen presentation. In line with human macrophages [25], this phenomenon is attributed to the *Brucella* Omp25 protein-induced down-regulation of TNFα production from infected DCs [32]. In addition, infected human DCs are poor inducers of naive T-lymphocyte proliferation, due to poor production of IL-12, which is essential in driving Th1 immune responses [32]. In the above model, S-LPS strains arrest

the maturation of DCs in comparison to rough mutants. However, this effect is not related to purified S-LPS, indicating an indirect involvement of the O-side chain (OPS) [31].

On the contrary, others have demonstrated that *Brucella* Omp19 lipoprotein can efficiently stimulate human monocyte-derived DCs' phenotypic and functional maturation, in a TLR2 and TLR4 dependent manner, providing a mechanism by which these bacteria generate Th1-type immune responses [33]. Probably, these conflicting results are related to the differences of the various *in vitro* systems used. Similar discrepancies have also been reported for another intracellular pathogen, *Mycobacterium tuberculosis*. It has been speculated that at the onset of infection *Brucella* activates DCs and induces Th1 responses, but at later time it could circumvent these Th1 responses to establish a chronic infection by means of different evasion mechanisms such as down-regulation of MHC-II expression in macrophages [26] or the inhibition of DC maturation [31,32].

Regarding *Brucella* virB operon, although it plays an important role in infected macrophages [18] it seems that it is not implicated in the control of DC maturation and does not interfere with the establishment of Th1 specific immune responses [34].

3. Immune deviation in brucellosis patients

The *in vitro* experiments in primary human cells or human cell lines described above, clearly indicate that *Brucella* modifies the function of professional antigen-presenting cells and circumvents the innate immune mechanisms to favor its survival (Table 1). Actually, chemotaxis in response to *Brucella* antigen and phagocytosis have been found to be significantly down-regulated in polymorphonuclear and mononuclear phagocytes of active brucellosis patients. The alterations in phagocytic cell function were more pronounced in patients with bacteremia, with focal manifestations or with longer disease duration [35]. A deficient phagocytic activity of peripheral blood monocytes in acute brucellosis patients has also been recently reported [36]. Moreover, defective antioxidant defense in terms of reduced leukocyte superoxide dismutase (SOD) activity has been described in acute brucellosis patients [37]. Successful treatment of acute disease resulted to complete or partial revert of the phagocytic capacity and increase of SOD enzyme activity [35–37].

On the other hand, in the chronic disease stage patient monocytes are characterized by persisting phagocytosis disturbance as well as random and directed migration against non-specific and disease specific antigens [38,39]. Other studies have described anergy in chronic brucellosis patients, in the context of significant decrease to leukocyte inhibition migration factor and decline in skin reactivity to different antigens [40–43].

In vitro studies have demonstrated that *Brucella* inhibits spontaneously occurring apoptosis in both invaded and non-invaded human monocytes indicating the existence of potential anti-apoptotic factors induced during infection. Ex vivo studies revealed that monocytes and lymphocytes from brucellosis patients were highly resistant to spontaneously or Fas-mediated apoptosis compared to healthy controls [44]. Of note, monocytes and lymphocytes from chronically infected patients, in contrast to acutely ill patients, remained resistant to apoptosis despite proper antimicrobial treatment [44].

Th1 immune response against *Brucella* leads to IFNγ and IL-2 secretion by antigen specific CD4+ and CD8+ T-lymphocytes. In general, IFNγ strongly activates the bactericidal function of macrophages, stimulates CD8+ T-lymphocyte mediated-cytotoxicity and potentiates apoptotic death of infected macrophages. Additionally, IFNγ and IL-2 promote the expression of antigen presenting and costimulatory molecules on DCs/macrophages. IL-2 is a growth factor for antigen-stimulated T-lymphocytes and is responsible for the Th1 clonal expansion [8].

Given that the innate immune system controls activation of the Th1 adaptive immunity and *Brucella* subverts innate immune mechanisms, it was expected that chronic brucellosis patients would display disturbed Th1 immune response and T-cell anergy. Depressed T-cell immunity in chronic brucellosis patients has been acknowledged for long [45,46]. Numerous quantitative and qualitative alterations of peripheral T-lymphocytes have been observed in chronically suffering patients through the years. Generally, an increase of cytotoxic CD8+ T-lymphocytes has been reported in persistent brucellosis or during relapses [47,48], possibly compensating for an ineffectual CD4+ T-cell response, at least in chronic brucellosis patients.

PBMCs from chronic brucellosis patients showed low proliferative response to non-specific (mitogens) or specific *Brucella* antigens, in comparison to acute brucellosis patients [49,50]. Recently, low proliferative response to *Brucella* cell extract antigen in comparison to PHA and PPD has also been demonstrated in PBMCs from patients with chronic brucellosis [51]. More specifically, suppressed T-cell proliferative response to PHA in chronic brucellosis patients was related to CD4+ subset [52]. In another study, T-lymphocytes from chronic brucellosis patients displayed reduced expression of early activation marker, CD69, although low proliferative response was not demonstrated [47].

The expression of high affinity IL-2 receptor (CD25) and CD80/CD28 costimulation molecules on PHA-cultured PBMCs in human brucellosis patients has also been studied [4,48,53]. Interestingly, chronic relapsing brucellosis patients present diminished percentage of peripheral CD4+ T-lymphocytes expressing CD25 when compared to acutely ill patients, while retaining low frequency of CD4+/CD25+ and CD4+/CD28+ T-cell subsets even after potent *in vitro* stimulation of PBMCs with PHA and *E coli* LPS [48,53]. This phenomenon may be attributed to the immunosuppressive effect of *Brucella* antigens on patients' PBMCs, given that HKBA stimulation of PHA-cultured PBMCs from brucellosis patients led to a significant reduction of CD4+/CD25+ T-cells and CD80 + monocytes in a dose-related manner [4,53].

Several studies have reported serum or intracytoplasmic overproduction of Th1 cytokines (IFNγ mainly) in acute brucellosis that were reduced or normalized after adequate antibiotic treatment [36,54,55]. Pre-treatment CD3+IFNγ+ T-lymphocytes levels were found significantly lower in patients who did not respond to treatment indicating a probable association between defective Th1 responses and chronic disease stage [55].

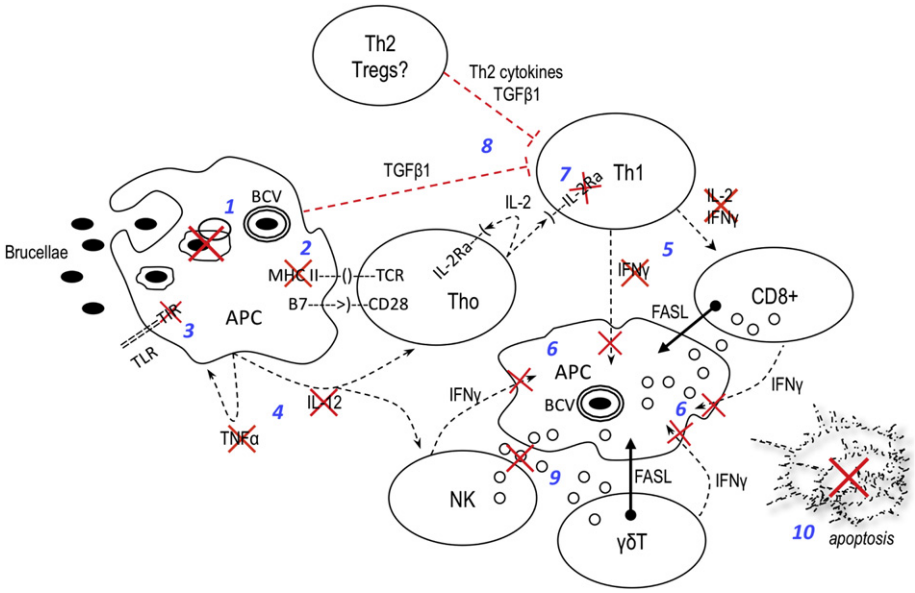


Fig. 1. Proposed effects of chronic *Brucella* infection on human cellular immune mechanisms leading to defective Th1 immune response and anergy. Red lines indicate negative effect. 1) Inhibition of phagolysosome fusion and promotion of BCV formation, 2) Down modulation of antigen presentation, 3) Interfere with TLR signaling, 4) Blockade of TNFα and IL-12 production, 5) Diminished production of IL-2 and IFNγ, 6) Interfere with IFNγ signaling, 7) Low frequency of IL-2Ra positive Th cells, 8) Production of Th2 cytokines (eg IL-10, IL-13) and TGFβ1, 9) Deficient NK cytotoxicity, 10) Prevention of apoptosis. APC = Antigen-Presenting Cell, BCV = *Brucella* Containing Vacuoles, IL-2Ra = IL-2 receptor a, Tho = naïve T-helper cells, Tregs = T regulatory cells, FASL = Fas Ligand. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Noteworthy, reduced *in vitro* T-cell blastogenesis in patients suffering from chronic brucellosis was linked to cytokine network imbalance [49–51]. More specifically, PBMCs from chronic anergic brucellosis patients display significantly low levels of IFNγ production when stimulated with *Brucella* protein [49]. In addition, HKBA-cultured whole blood from patients with prolonged disease was characterized by reduced IFNγ production, significant reduction of CD3+IFNγ+ T-cells and increment of CD3+IL-13+ T-cells [50]. Recently, it has also been suggested that unresponsiveness to brucellar antigens is mediated by TGFβ1 in chronic brucellosis patients [51]. These results imply that at the onset of human brucellosis a Th1 response dominates which is diminishing during disease prolongation probably due to Th2 switch and/or release of immunoregulative cytokines such as TGFβ1.

Table 2 summarizes T-cell immunity disturbances that have been reported in chronic persistent/relapsing brucellosis patients.

Apart from αβ TCR CD4+ and CD8+ T-lymphocytes, natural killer (NK) cells and rare, non-conventional, T-lymphocytes, such as Vγ9Vδ2 T cells and CD4+ invariant NKT cells (iNKT) seem to contribute to antibacterial immunity against *Brucella* [56–58]. In particular, human *in vitro* studies have demonstrated that they impair *Brucella* intra-macrophagic growth through a combination of contact dependent and independent mechanisms including granule- (NK cells, Vγ9Vδ2 T-cells, CD4+ iNKT cells) and Fas ligand- (Vγ9Vδ2 T-cells, CD4+ iNKT cells) mediated-cytotoxicity, activation of macrophages by IFNγ production (NK cells, Vγ9Vδ2 T-cells, CD4+ iNKT cells) and secretion of the potent bactericidal factor LL-37 cathelicidin (Vγ9Vδ2 T-cells) [56–58]. Interestingly, peripheral blood Vγ9Vδ2 T-cells increase significantly in acute brucellosis patients and decline after successful treatment [59]. On the other hand, a deficient NK cytotoxicity, which reverts to normal after antibiotic treatment, has been reported in the PBMCs of acute brucellosis patients, suggesting that the efficiency of NK response could be important for the evolution of the disease in humans [60].

Substantial evidence exists that host genes are important determinants of susceptibility to or outcome of infectious diseases. During the last decade, a variety of human genetic studies were designed to associate susceptibility to brucellosis and development of focal or chronic disease with polymorphisms or mutations of molecules (mainly cytokines) involved in immune reactions (Table 3) [61–83]. However, in our opinion the results should be considered with reservation because of the limited population size, the clinical samples' heterogeneity and the absence of meta-analyses.

Taking into account depressed cellular immunity in chronic anergic/relapsing brucellosis patients, the immunostimulatory effect of levamisole, ascorbic acid, IFNα and bacterial extracts administered as a supplement to the conventional antibiotic treatment seems to be beneficial in reconstituting cellular immunity responses leading to chronicity recovery in many cases [38–43,45,84,85]. However, conflicting results have been reported [86].

4. Conclusions

Human protection against *Brucella* depends on cell-mediated immunity, involving mainly activated antigen-presenting cells (macrophages, dendritic cells) and CD4+, CD8+ T-lymphocytes. Other cellular populations such as NK cells and non-conventional T-lymphocytes, like Vγ9Vδ2 T cells and CD4+ invariant NKT cells (iNKT), could also contribute to cellular immunity against *Brucella*. Th1 immune response

dominates at the onset of brucellosis and is essential for the clearance of infection. On the other hand, *Brucella* successfully survives and replicates inside antigen-presenting cells by remodeling the intracellular compartment. For this purpose *Brucella* produces various virulence factors that modify phagocytic capacity, antigen presentation, cellular maturation, cytokine secretion, and apoptosis. Modulation of innate immune mechanisms by *Brucella* leads to the depression of T-cell immunity in chronic brucellosis patients, characterized by defective Th1 immune responses and T-cell anergy (Fig. 1). Human immunogenetic profile may correlate with disease susceptibility or protection, although larger studies and meta-analyses should be carried out.

Further elucidation of human brucellosis immunopathogenesis is important for the discovery of novel diagnostic/predictive markers concerning disease relapse and chronicity, as well as development of new therapeutic approaches and vaccination strategies in livestock and humans.

References

[1] G. Pappas, N. Akritidis, M. Bosilkovski, E. Tsianos, Brucellosis, N. Engl. J. Med. 352 (2005) 2325–2336.

[2] G. Pappas, P. Panagopoulou, L. Christou, N. Akritidis, *Brucella* as a biological weapon, Cell. Mol. Life Sci. 63 (2006) 2229–2236.

[3] E. Bouza, C. Sánchez-Carrillo, S. Hernangómez, M.J. González, Spanish Co-operative Group for the study of laboratory-acquired brucellosis: a Spanish national survey, J. Hosp. Infect. 61 (2005) 80–83.

[4] P. Skendros, P. Boura, F. Kamaria, M. Raptopoulou-Gigi, CD80/CD28 co-stimulation in human brucellosis, Clin. Exp. Immunol. 146 (2006) 400–408.

[5] B. Golding, D.E. Scott, O. Scharf, L.Y. Huang, M. Zaitseva, C. Lapham, N. Eller, H. Golding, Immunity and protection against *Brucella abortus*, Microbes Infect. 3 (2001) 43–48.

[6] C.L. Baldwin, R. Goenka, Host immune responses to the intracellular bacteria *Brucella*: does the bacteria instruct the host to facilitate chronic infection? Crit. Rev. Immunol. 26 (2006) 407–442.

[7] J.P. Gorvel, Brucella: a Mr "Hide" converted into Dr Jekyll, Microbes Infect. 10 (2008) 1010–1013.

[8] S. Yingst, D.L. Hoover, T cell immunity to brucellosis, Crit. Rev. Microbiol. 29 (2003) 313–331.

[9] M.P. Jimenez de Bagues, S. Dudal, J. Dornand, A. Gross, Cellular bioterrorism: how *Brucella* corrupts macrophage physiology to promote invasion and proliferation, Clin. Immunol. 114 (2005) 227–238.

[10] A. Gross, S. Bertholet, J. Mauel, J. Dornand, Impairment of *Brucella* growth in human macrophagic cells that produce nitric oxide, Microb. Pathog. 36 (2004) 75–82.

[11] M. Zaitseva, L.R. King, J. Manischewitz, M. Dougan, L. Stevan, H. Golding, B. Golding, Human peripheral blood T cells, monocytes, and macrophages secrete macrophage inflammatory proteins 1alpha and 1beta following stimulation with heat-inactivated *Brucella abortus*, Infect. Immun. 69 (2001) 3817–3826.

[12] M.G. Rittig, A. Kaufmann, A. Robins, B. Shaw, H. Sprenger, D. Gemsa, V. Foulongne, B. Rouot, J. Dornand, Smooth and rough lipopolysaccharide phenotypes of *Brucella* induce different intracellular trafficking and cytokine/chemokine release in human monocytes, J. Leukoc. Biol. 74 (2003) 1045–1055.

[13] E. Caron, A. Gross, J.P. Liautard, J. Dornand, *Brucella* species release a specific, protease-sensitive, inhibitor of TNF-alpha expression, active on human macrophage-like cells, J. Immunol. 156 (1996) 2885–2893.

[14] C.E. Papagoras, M.I. Argyropoulou, P.V. Voulgari, I. Vrabie, A.K. Zikou, A.A. Drosos, A case of *Brucella spondylitis* in a patient with psoriatic arthritis receiving infliximab, Clin. Exp. Rheumatol. 27 (2009) 124–127.

[15] G. Vrioni, G. Pappas, E. Priavali, C. Gartzonika, S. Levidiotou, An eternal microbe: *Brucella* DNA load persists for years after clinical cure, Clin. Infect. Dis. 46 (2008) e131–136.

[16] M.G. Rittig, M.T. Alvarez-Martinez, F. Porte, J.P. Liautard, B. Rouot, Intracellular survival of *Brucella* spp. in human monocytes involves conventional uptake but special phagosomes, Infect. Immun. 69 (2001) 3995–4006.

[17] B.H. Bellaire, R.M. Roop 2nd, J.A. Cardelli, Opsonized virulent *Brucella abortus* replicates within nonacidic, endoplasmic reticulum-negative, LAMP-1-positive phagosomes in human monocytes, Infect. Immun. 73 (2005) 3702–3713.

[18] M.L. Boschiroli, S. Ouahrani-Bettache, V. Foulongne, S. Michaux-Charachon, G. Bourg, A. Allardet-Servent, C. Cazevieille, J.P. Liautard, M. Ramuz, D. O'Callaghan, The *Brucella suis* virB operon is induced intracellularly in macrophages, Proc. Natl. Acad. Sci. U S A 99 (2002) 1544–1549.

[19] J.P. Lavigne, G. Patey, F.J. Sangari, G. Bourg, M. Ramuz, D. O'Callaghan, S. Michaux-Charachon, Identification of a new virulence factor, BvfA, in *Brucella suis*, Infect. Immun. 73 (2005) 5524–5529.

[20] P.G. Cardoso, G.C. Macedo, V. Azevedo, S.C. Oliveira, *Brucella* spp noncanonical LPS: structure, biosynthesis, and interaction with host immune system, Microb. Cell Fact. 5 (2006) 13.

[21] C.M. Fernandez-Prada, E.B. Zelazowska, M. Nikolich, T.L. Hadfield, R.M. Roop 2nd, G.L. Robertson, D.L. Hoover, Interactions between *Brucella melitensis* and human phagocytes: bacterial surface O Polysaccharide inhibits phagocytosis, bacterial killing, and subsequent host cell apoptosis, Infect. Immun. 71 (2003) 2110–2119.

[22] C.M. Fernandez-Prada, M. Nikolich, R. Vemulapalli, N. Sriranganathan, S.M. Boyle, G.G. Schurig, T.L. Hadfield, D.L. Hoover, Deletion of wboA enhances activation of the lectin pathway of complement in *Brucella abortus* and *Brucella melitensis*, Infect. Immun. 69 (2001) 4407–4416.

[23] A. Kariminia, G. Kavoossy, S. Khatami, E. Zowghi, S.K. Ardestani, Study of interleukin-10 and interleukin-12 productions in response to lipopolysaccharides extracted from two different *Brucella* strains, Comp. Immunol. Microbiol. Infect. Dis. 25 (2002) 85–93.

[24] G.H. Giambartolomei, A. Zwerdling, J. Cassataro, L. Bruno, C.A. Fossati, M.T. Philipp, Lipoproteins, not lipopolysaccharide, are the key mediators of the proinflammatory response elicited by heat-killed *Brucella abortus*, J. Immunol. 173 (2004) 4635–4642.

[25] V. Jubier-Maurin, R.A. Boigegrain, A. Cloeckaert, A. Gross, M.T. Alvarez-Martinez, A. Terraza, J. Liautard, S. Köhler, B. Rouot, J. Dornand, J.P. Liautard, Major outer membrane protein Omp25 of *Brucella suis* is involved in inhibition of tumor necrosis factor alpha production during infection of human macrophages, Infect. Immun. 69 (2001) 4823–4830.

[26] P. Barrionuevo, J. Cassataro, M.V. Delpino, A. Zwerdling, K.A. Pasquevich, C. García Samartino, J.C. Wallach, C.A. Fossati, G.H. Giambartolomei, *Brucella abortus* inhibits major histocompatibility complex class II expression and antigen processing through interleukin-6 secretion via Toll-like receptor 2, Infect. Immun. 76 (2008) 250–262.

[27] G.K. Radhakrishnan, Q. Yu, J.S. Harms, G.A. Splitter, *Brucella* TIR Domain-containing protein mimics properties of the toll-like receptor adaptor protein TIRAP, J. Biol. Chem. 284 (2009) 9892–9898.

[28] D. Sengupta, A. Koblansky, J. Gaines, T. Brown, A.P. West, D. Zhang, T. Nishikawa, S.G. Park, R.M. Roop 2nd, S. Ghosh, Subversion of innate immune responses by *Brucella* through the targeted degradation of the TLR signaling adapter, MAL, J. Immunol. 184 (2010) 956–964.

[29] S. Bouhet, V. Lafont, E. Billard, A. Gross, J. Dornand, The IFN gamma-induced STAT1-CBP/P300 association, required for a normal response to the cytokine, is disrupted in *Brucella*-infected macrophages, Microb. Pathog. 46 (2009) 88–97.

[30] A. Gross, M. Bouaboula, P. Casellas, J.P. Liautard, J. Dornand, Subversion and utilization of the host cell cyclic adenosine 5'-monophosphate/protein kinase A pathway by *Brucella* during macrophage infection, J. Immunol. 170 (2003) 5607–5614.

[31] E. Billard, J. Dornand, A. Gross, Interaction of *Brucella suis* and *Brucella abortus* rough strains with human dendritic cells, Infect. Immun. 75 (2007) 5916–5923.

[32] E. Billard, J. Dornand, A. Gross, *Brucella suis* prevents human dendritic cell maturation and antigen presentation through regulation of tumor necrosis factor alpha secretion, Infect. Immun. 75 (2007) 4980–4989.
[33] A. Zwerdling, M.V. Delpino, P. Barrionuevo, J. Cassataro, K.A. Pasquevich, C. García Samartino, C.A. Fossati, G.H. Giambartolomei, *Brucella* lipoproteins mimic dendritic cell maturation induced by *Brucella abortus*, Microbes Infect. 10 (2008) 1346–1354.
[34] E. Billard, J. Dornand, A. Gross, VirB type IV secretory system does not contribute to *Brucella suis*' avoidance of human dendritic cell maturation, FEMS Immunol. Med. Microbiol. 53 (2008) 404–412.
[35] P. Ocon, J.M. Reguera, P. Morata, C. Juarez, A. Alonso, J.D. Colmenero, Phagocytic cell function in active brucellosis, Infect. Immun. 62 (1994) 910–914.
[36] M. Rodríguez-Zapata, M.J. Matías, A. Prieto, M.A. Jonde, J. Monserrat, L. Sánchez, E. Reyes, A. De la Hera, M. Alvarez-Mon, Human brucellosis is characterized by an intense Th1 profile associated to a defective monocyte function, Infect. Immun. 78 (2010) 3272–3279.
[37] A.B. Karabulut, E. Sonmez, Y. Bayindir, Effect of the treatment of brucellosis on leukocyte superoxide dismutase activity and plasma nitric oxide level, Ann. Clin. Biochem. 42 (2005) 130–132.
[38] P. Boura, M. Raptopoulou-Gigi, G. Goulis, Peripheral blood monocyte phagocytosis in patients with chronic brucellosis, J. Immunopharmacol. 4 (1982–1983) 377–381.
[39] P. Boura, G. Tsapas, A. Papadopoulou, I. Magoula, G. Kountouras, Monocyte locomotion in anergic chronic brucellosis patients: the in vivo effect of ascorbic acid, Immunopharmacol. Immunotoxicol. 11 (1989) 119–129.
[40] F. Lagra, M. Raptopoulou-Gigi, H. Orphanou-Koumerkeridou, G. Goulis, In vitro effect of interferon alpha-2b on T lymphocyte transformation and leukocyte migration inhibition in patients with chronic brucellosis, Immunopharmacol. Immunotoxicol. 11 (1989) 223–232.
[41] S. Printzis, M. Raptopoulou-Gigi, H. Orphanou-Koumerkeridou, F. Lagre, G. Goulis, Immunotherapy in chronic brucellosis. Effect of levamisole and interferon; mechanisms of action and clinical value, Immunopharmacol. Immunotoxicol. 16 (1994) 679–693.
[42] P. Boura, G. Tsapas, J. Kountouras, E. Zaharioudaki, Interferon-A 2a administration in chronic anergic brucellosis, Hepatogastroenterology 42 (1995) 919–922.
[43] P. Boura, P. Skendros, J. Kountouras, E. Zacharioudaki, T. Tsapas, Effect of bacterial extracts on the immunologic profile in chronic relapsing brucellosis patients, Int. J. Immunopathol. Pharmacol. 12 (1999) 103–111.
[44] M. Tolomeo, P. Di Carlo, V. Abbadessa, L. Titone, S. Miceli, E. Barbusca, G. Cannizzo, S. Mancuso, S. Arista, F. Scarlata, Monocyte and lymphocyte apoptosis resistance in acute and chronic brucellosis and its possible implications in clinical management, Clin. Infect. Dis. 36 (2003) 1533–1538.
[45] M. Renoux, G. Renoux, Brucellosis, immunodepression, and levamisole, Lancet 1 (1977) 372.
[46] R.D. Thornes, Acquired immune suppression in chronic brucellosis, Ir. Med. J. 76 (1983) 225.
[47] M.C. Moreno-Lafont, R. Lopez-Santiago, E. Zumaran-Cuellar, V. Paredes-Cervantes, A. López-Merino, A. Estrada-Aguilera, L. Santos-Argumedo, Antigen-specific activation and proliferation of CD4+ and CD8+ T lymphocytes from brucellosis patients, Trans. R. Soc. Trop. Med. Hyg. 96 (2002) 340–347.
[48] P. Skendros, P. Boura, D. Chrisagis, M. Raptopoulou-Gigi, Diminished percentage of CD4+ T-lymphocytes expressing interleukine-2 receptor alpha in chronic brucellosis, J. Infect. 54 (2007) 192–197.
[49] G.H. Giambartolomei, M.V. Delpino, M.E. Cahanovich, J.C. Wallach, P.C. Baldi, C.A. Velikovsky, C.A. Fossati, Diminished production of T helper 1 cytokines correlates with T cell unresponsiveness to *Brucella* cytoplasmic proteins in chronic human brucellosis, J. Infect. Dis. 186 (2002) 252–259.
[50] A. Rafiei, S.K. Ardestani, A. Kariminia, A. Keyhani, M. Mohraz, A. Amirkhani, Dominant Th1 cytokine production in early onset of human brucellosis followed by switching towards Th2 along prolongation of disease, J. Infect. 53 (2006) 315–324.
[51] M.G. Elfaki, A.A. Al-Hokail, Transforming growth factor beta production correlates with depressed lymphocytes function in humans with chronic brucellosis, Microbes Infect. 11 (2009) 1089–1096.
[52] S. Kinikli, N. Turkcapar, M.B. Kucukay, G. Keskin, G. Kinikli, In vitro nonspecific mitogenic response of T-cell subsets in acute and chronic brucellosis, Diagn. Microbiol. Infect. Dis. 52 (2005) 229–233.
[53] P. Skendros, A. Sarantopoulos, K. Tselios, P. Boura, Chronic brucellosis patients retain low frequency of CD4+ T-lymphocytes expressing CD25 and CD28 after Escherichia coli LPS stimulation of PHA-cultured PBMCs, Clin. Dev. Immunol. (2008) 327346.
[54] K. Demirdag, M. Ozden, A. Kalkan, A. Godekmerdan, S. Sirri Kilic, Serum cytokine levels in patients with acute brucellosis and their relation to the traditional inflammatory markers, FEMS Immunol. Med. Microbiol. 39 (2003) 149–153.
[55] H.H. Akbulut, S.S. Kilic, V. Bulut, M. Ozden, Determination of intracellular cytokines produced by Th1 and Th2 cells using flow cytometry in patients with brucellosis, FEMS Immunol. Med. Microbiol. 45 (2005) 253–258.
[56] J. Dornand, V. Lafont, J. Oliaro, A. Terraza, E. Castaneda-Roldan, J.P. Liautard, Impairment of intramacrophagic *Brucella suis* multiplication by human natural killer cells through a contact-dependent mechanism, Infect. Immun. 72 (2004) 2303–2311.
[57] S. Dudal, C. Turriere, S. Bessoles, P. Fontes, F. Sanchez, J. Liautard, J.P. Liautard, V. Lafont, Release of LL-37 by activated human Vgamma9V-delta2 T cells: a microbicidal weapon against *Brucella suis*, J. Immunol. 177 (2006) 5533–5539.
[58] S. Bessoles, S. Dudal, G.S. Besra, F. Sanchez, V. Lafont, Human CD4+ invariant NKT cells are involved in antibacterial immunity against *Brucella suis* through CD1d-dependent but CD4-independent mechanisms, Eur. J. Immunol. 39 (2009) 1025–1035.
[59] S.S. Kilic, H.H. Akbulut, M. Ozden, V. Bulut, Gamma/delta T cells in patients with acute brucellosis, Clin. Exp. Med. 9 (2009) 101–104.
[60] I. Salmerón, M. Rodríguez-Zapata, O. Salmerón, L. Manzano, S. Vaquer, M. Alvarez-Mon, Impaired activity of natural killer cells in patients with acute brucellosis, Clin. Infect. Dis. 15 (1992) 764–770.
[61] A. Caballero, M.J. Bravo, A. Nieto, J.D. Colmenero, A. Alonso, J. Martin, TNFA promoter polymorphism and susceptibility to brucellosis, Clin. Exp. Immunol. 121 (2000) 480–483.
[62] P. Skendros, P. Boura, N. Tsantas, P. Debre, I. Theodorou, Frequency analysis of the CCR5delta32 mutation in patients with brucellosis, Scand. J. Infect. Dis. 34 (2002) 944–946.
[63] G. Orozco, E. Sánchez, M.A. López-Nevot, A. Caballero, M.J. Bravo, P. Morata, J. de Dios Colmenero, A. Alonso, J. Martín, Inducible nitric oxide synthase promoter polymorphism in human brucellosis, Microbes Infect. 5 (2003) 1165–1169.
[64] M.J. Bravo, J. de Dios Colmenero, A. Alonso, A. Caballero, Polymorphisms of the interferon gamma and interleukin 10 genes in human brucellosis, Eur. J. Immunogenet. 30 (2003) 433–435.
[65] M.J. Bravo, J. de Dios Colmenero, A. Alonso, A. Caballero, HLA-B*39 allele confers susceptibility to osteoarticular complications in human brucellosis, J. Rheumatol. 30 (2003) 1051–1053.
[66] M.J. Bravo, J. de D. Colmenero, J. Martín, A. Alonso, A. Caballero González, Variation in the NRAMP1 gene does not affect susceptibility or protection in human brucellosis, Microbes Infect. 8 (2006) 154–156.
[67] M. Rezazadeh, M. Hajilooi, A. Rafiei, M. Haidari, E. Nikoopour, F. Kerammat, M. Mamani, M. Ranjbar, H. Hashemi, TLR4 polymorphism in Iranian patients with brucellosis, J. Infect. 53 (2006) 206–210.
[68] M. Rezazadeh, M. Hajilooi, M. Haidari, A. Rafiei, S.A. Alavi, F. Keramat, Association of susceptibility to brucellosis and interleukin-4 promoter polymorphism, Scand. J. Infect. Dis. 38 (2006) 1045–1049.
[69] M. Haidari, M. Hajilooi, M. Rezazadeh, A. Rafiei, S.A. Alavi, F. Keramat, Polymorphism in the promoter region of the CD14 gene and susceptibility to Brucellosis, Immunol. Invest. 35 (2006) 239–245.
[70] A. Rafiei, M. Hajilooi, R.J. Shakib, S. Shams, N. Sheikh, Association between the Phe206Leu polymorphism of L-selectin and brucellosis, J. Med. Microbiol. 55 (2006) 511–516.
[71] M. Hajilooi, A. Rafiei, M. Reza Zadeh, N. Tajik, Association of interleukin-1 receptor antagonist gene polymorphism and susceptibility to human brucellosis, Tissue Antigens 68 (2006) 331–334.
[72] S. Davoudi, A.A. Amirzargar, M. Hajiabdolbaghi, M. Rasoolinejad, A. Soodbakhsh, S. Jafari, H. Piri, P. Maleknejad, H. Bagherian, N. Madadi,

B. Nikbin, Th-1 cytokines gene polymorphism in human brucellosis, Int. J. Immunogenet. 33 (2006) 355−359.
[73] A. Rafiei, M. Hajilooi, R.J. Shakib, S.A. Alavi, Transforming growth factor-beta1 polymorphisms in patients with brucellosis: an association between codon 10 and 25 polymorphisms and brucellosis, Clin. Microbiol. Infect. 13 (2007) 97−100.
[74] F. Budak, G. Göral, Y. Heper, E. Yilmaz, F. Aymak, B. Baştürk, O. Töre, B. Ener, H.B. Oral, IL-10 and IL-6 gene polymorphisms as potential host susceptibility factors in Brucellosis, Cytokine 38 (2007) 32−36.
[75] M. Rasouli, S. Kiany, Association of interferon-gamma and interleukin-4 gene polymorphisms with susceptibility to brucellosis in Iranian patients, Cytokine 38 (2007) 49−53.
[76] M.J. Bravo, J.D. Colmenero, J. Martín, A. Alonso, A. Caballero, Polymorphism of the transmembrane region of the MICA gene and human brucellosis, Tissue Antigens 69 (2007) 358−360.
[77] A. Rafiei, M. Hajilooi, M. Vahedi, R.J. Shakib, The Ser128Arg polymorphism for E-selectin gene and brucellosis, Infect. Genet. Evol. 7 (2007) 494−498.
[78] S.H. Hashemi, M. Hajilooi, M. Mamani, S. Jamal-Omidi, Fc gamma receptor IIa polymorphism in patients with brucellosis, Jpn. J. Infect. Dis. 60 (2007) 196−197.
[79] M. Rasouli, S. Kiany, M. Behbin, Interleukin-10 gene polymorphisms and susceptibility to brucellosis in Iranian patients, Iran J. Immunol. 5 (2008) 131−135.
[80] M.J. Bravo, J.D. Colmenero, M.I. Queipo-Ortuño, A. Alonso, A. Caballero, TGF-beta1 and IL-6 gene polymorphism in Spanish brucellosis patients, Cytokine 44 (2008) 18−21.
[81] I. Karaoglan, S. Pehlivan, M. Namiduru, M. Pehlivan, C. Kilinçarslan, Y. Balkan, I. Baydar, TNF-alpha, TGF-beta, IL-10, IL-6 and IFN-gamma gene polymorphisms as risk factors for brucellosis, New Microbiol. 32 (2009) 173−178.
[82] M.J. Bravo, J.D. Colmenero, M.I. Queipo-Ortuño, P. Morata, G. Orozco, J. Martin, A. Alonso, A. Caballero, PTPN22 C1858T polymorphism and human brucellosis, Scand. J. Infect. Dis. 41 (2009) 109−112.
[83] M.J. Bravo, J.D. Colmenero, M.I. Queipo-Ortuño, J. Martín, R. Lavado, A. Alonso, A. Caballero, No association of TAP and LMP genetic polymorphism in human brucellosis and its complications, Hum. Immunol. 71 (2010) 708−711.
[84] M. Raptopoulou-Gigi, P. Boura, N. Valcanos, G. Goulis, Conventional therapy and levamisole in the management of chronic brucellosis, J. Infect. Dis. 147 (1983) 1123−1124.
[85] H. Irmak, T. Buzgan, M.K. Karahocagil, O. Evirgen, H. Akdeniz, A.P. Demiröz, The effect of levamisole combined with the classical treatment in chronic brucellosis, Tohoku J. Exp. Med. 201 (2003) 221−228.
[86] U. Dizer, L. Hayat, C.M. Beker, L. Görenek, V. Ozgüven, A. Pahsa, The effect of the doxycycline-rifampicin and levamisole combination on lymphocyte subgroups and functions of phagocytic cells in patients with chronic brucellosis, Chemotherapy 51 (2005) 27−31.